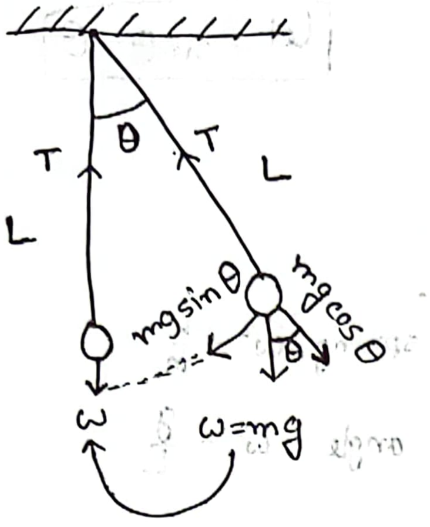
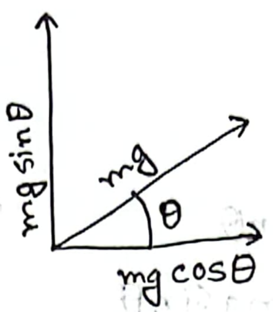
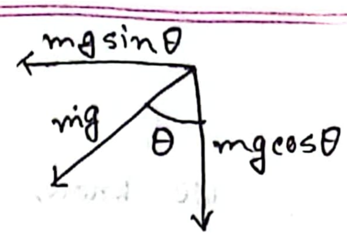






θ যেদিকে কোণ উৎপন্ন করে সে বরাবর cosθ হবে।

Tension always thread (সুতা) বরাবর কাজ করে। Tension force and mgcosθ একে অপরের সম্পূর্ণ বিপরীত দিকে কাজ করে তাই vanish হয়ে যায়। শুধু mgsinθ থাকবে।

$T = mg\cos\theta$ বিপরীত ও সমান।

$$F_{restore} = -mg\sin\theta$$
$$= -mg\theta$$

θ এর মান যদি অনেক ছোট হয় তাহলে $\sin\theta \simeq \theta$

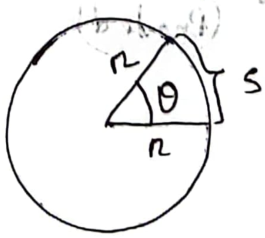
$x = L\theta$

$$\Rightarrow \frac{d^2x}{dt^2} = L\frac{d^2\theta}{dt^2} \quad [a \text{ এর জন্য double diff}]$$

$$\Rightarrow a = L\frac{d^2\theta}{dt^2}$$

θ = angular displacement

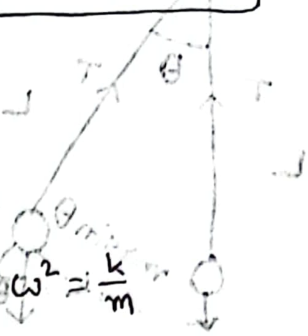
x = Linear displacement



$\therefore S = r\theta$

$x = L\theta$

$$\boxed{\text{let } \sin\theta = \theta}$$



we know,

$$F = -mg\sin\theta \qquad a = L\frac{d^2\theta}{dt^2}$$

$$\Rightarrow ma = -mg\sin\theta$$

$$\Rightarrow ma = -mg\theta$$

$$\Rightarrow a = -g\theta$$

$$\Rightarrow L\frac{d^2\theta}{dt^2} = -g\theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\omega^2\theta$$

$$\boxed{\therefore \frac{d^2\theta}{dt^2} + \omega^2\theta = 0} \rightarrow \text{SHM}$$

in spring $\omega^2 = \frac{k}{m}$

in angle $\omega^2 = \frac{g}{L}$

in angular displacement,

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

in linear displacement

$$\frac{d^2\theta}{dt^2} + \omega^2 x = 0$$

now,

$$\omega^2 = \frac{g}{L}$$

$$\Rightarrow \omega = \sqrt{g/L}$$

$$\Rightarrow \frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{L}{g}}$$

(Proved)

16-oct

c-8

$T = 2\pi\sqrt{\frac{l}{g}}$

math-1

$T_1 = 2s,\ g_1 = 9.78\ ms^{-2}$ [L is constant]

$T_2 = 3s,\ g_2 = ?$

$$T_1 = 2\pi\sqrt{\frac{L}{g_1}} \quad \text{--- (i)}$$

$$T_2 = 2\pi\sqrt{\frac{L}{g_2}} \quad \text{--- (ii)}$$

(i) ÷ (ii)

$$\frac{T_1}{T_2} = \sqrt{\frac{g_2}{g_1}}$$

$$\Rightarrow g_2 = \frac{(T_1)^2}{(T_2)^2}\, g_1$$

Theory Question :

- Simple pendulum satisfies the differential equation of simple harmonic motion.
- Determination of the spring constant and effective mass of a given ~~spring~~ spiral spring

math-2

$T_1 = T,\ L_1 = L,\ L_2 = 2L,\ T_2 = ?$

next length will be doubled up of previous length.

g constant

$$T_1 = \sqrt{\frac{L_1}{g}},\quad T_2 = \sqrt{\frac{L_2}{g}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{L_1}{L_2}}$$

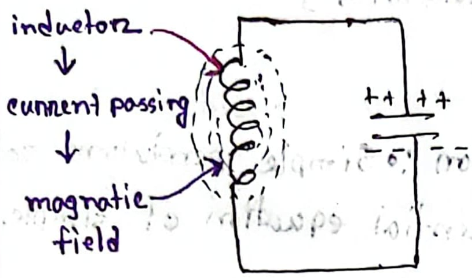
$$\Rightarrow T_2 = \sqrt{\frac{L_2}{L_1}} \times T_1$$

$$\Rightarrow T_2 = \sqrt{2L/L} \times T_1$$

$$\therefore T_2 = \sqrt{2} \times T_1$$

LC - Circuit

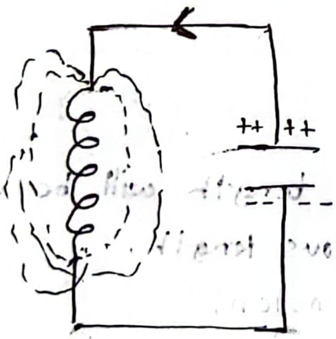
LC circuit : consists of inductor and capacitor



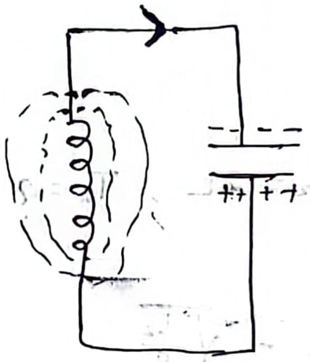
concept of magnatic force → current flow and vice versa

capacitor : can store charge
(current flow)

changing



dis changing

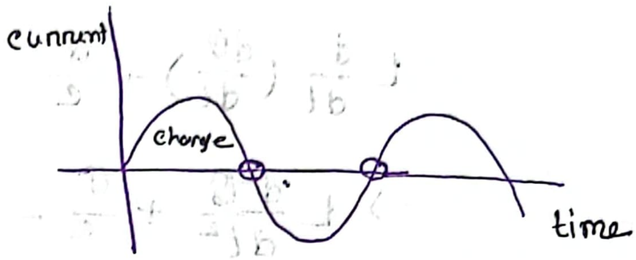


inductor থেকে যখন current flow হয় magnatic field creat হয়

circuit এ magnetic field তৈরি হলে capacitor এর charge শেষ হয়ে যায়। Then magnetic field capacitor কে charge করে, polarity change করে বা reverse হয়

RC circuit :

→ LC patern



Idal LC circuit
but এটা কে বাস্তবে পাওয়া যায় না because of $\boxed{\text{Resistance}}$

সূত্র

$$\boxed{Q = CV_c}$$

$$V_c = \frac{Q}{C}$$

V_c = Voltage capacitor
C = capacitor
Q = plate voltage

$$\boxed{V_L = L\frac{di}{dt}}$$

V_L = inductor এর দুই প্রান্তের / point এর voltage
i = current flow
$i = \frac{dQ}{dt}$

$V_L + V_c = 0$ [karshof law.]

$L\frac{di}{dt} + \frac{Q}{c} = 0$ $\left[i = \frac{dQ}{dt}\right]$

$L\frac{d}{dt}\left(\frac{dQ}{dt}\right) + \frac{Q}{c}$ $\left[\omega_0^2 = \frac{1}{Lc}\right]$

$\Rightarrow L\frac{d^2Q}{dt^2} + \frac{Q}{c} = 0$

$\Rightarrow \frac{d^2Q}{dt^2} + \frac{1}{Lc}Q = 0$

$\Rightarrow \frac{d^2Q}{dt^2} + \omega_0^2 Q = 0$ [so, charge motion SHM কে satisfy করে]

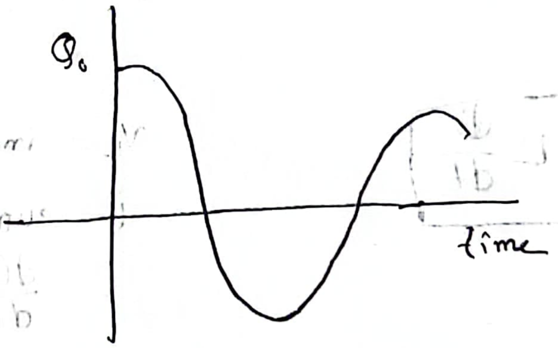
differential equation

what can be the possible solution of

$Q = Q_0 \sin\omega t$

$Q = Q_0 \cos\omega t$

chapter 908 page



∴ most sweetable solution is

$Q = Q_0 \cos\omega t$

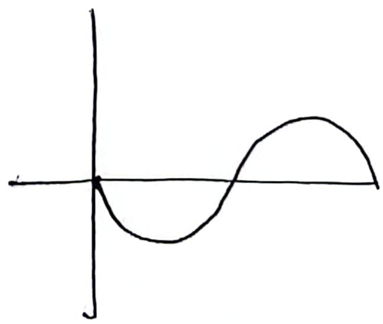
current flow :

$I = dQ/dt$

$= \frac{d}{dt}(Q_0 \cos\omega t)$

$= -\omega Q_0 \sin\omega t$ $[I_0 = \omega Q_0]$

$\boxed{= I = -I_0 \sin\omega t}$



\# $\omega^2 = \frac{1}{LC}$

$\Rightarrow \frac{2\pi}{T} = \frac{1}{\sqrt{LC}}$

$\Rightarrow T = 2\pi\sqrt{LC}$

$f = \frac{1}{2\pi\sqrt{LC}}$

L - inductance

C - capacitance

C-9/

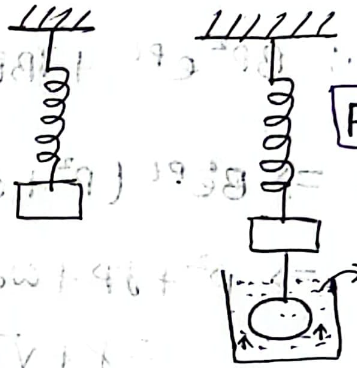
"Damped Oscillation"

~~F = ma = ...~~

$\boxed{F = ma = -kx - bv}$

→ hoks law

bv → damping force



$\boxed{F' \propto -v}$

$F' = -bv$

Resistivity দানে যেটা বাধা দিবে।

Damping force,

$F \propto -v$

$ma = -kx - bv$

$\boxed{b = \text{constant}}$

$\Rightarrow m\frac{d^2x}{dt^2} = -bv - kx$

$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}x - \frac{b}{m}\frac{dx}{dt}$

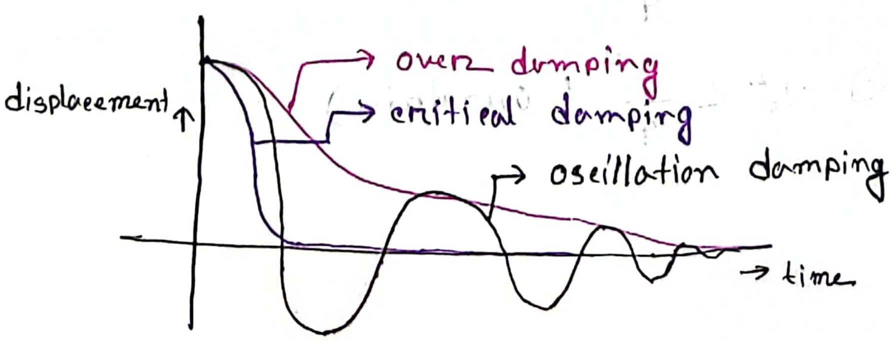
$\gamma = \frac{b}{m}$, $\omega^2 = \frac{k}{m}$

↳ damping constant

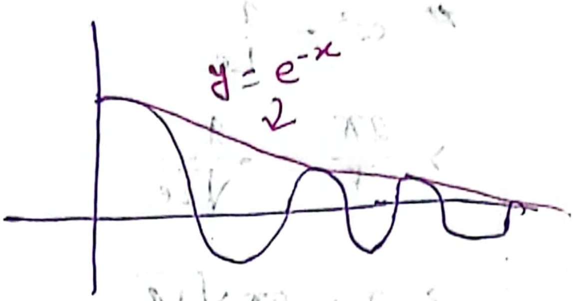
$\Rightarrow \frac{d^2x}{dt^2} = -\omega^2 x - \gamma\frac{dx}{dt}$

$$\boxed{\Rightarrow \frac{d^2x}{dt^2} + \gamma\frac{dx}{dt} + \omega^2 x = 0}$$

differential equation / damped HM doesn't satisfied SHM



$$\frac{d^2x}{dt^2} + \gamma\frac{dx}{dt} + \omega_0^2 x = 0$$



let, $x = Be^{pt}$

$$\frac{dx}{dt} = BPe^{pt}$$

$$\frac{d^2x}{dt^2} = BP^2 e^{pt}$$

$\therefore BP^2 e^{pt} + \gamma BPe^{pt} + \omega_0^2 Be^{pt} = 0$

$\Rightarrow Be^{pt}(P^2 + \gamma P + \omega_0^2) = 0$

$\Rightarrow P^2 + \gamma P + \omega_0^2 = 0$

$\therefore P = \dfrac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}$

$ax^2 + bx + c = 0$

$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\Rightarrow P = -\dfrac{\gamma}{2} \pm \sqrt{\dfrac{\gamma^2 - 4\omega_0^2}{4}}$

$\therefore P = -\dfrac{\gamma}{2} \pm \sqrt{\dfrac{\gamma^2}{4} - \omega_0^2}$ — (1)

from equation (1)

(1) $\omega_0^2 > \dfrac{\gamma^2}{4}$ (oscillatory)

(2) $\omega_0^2 = \dfrac{\gamma^2}{4}$ (critical)

(3) $\omega_0^2 < \dfrac{\gamma^2}{4}$ (overdrive)

Case 1:

$$\omega^2 > \frac{\gamma^2}{4}$$

$$= \sqrt{-(\omega_0^2 - \frac{\gamma^2}{4})} \qquad [\omega_1^2 = \omega_0^2 - \frac{\gamma}{4}]$$

$$= \sqrt{-\omega_1^2} \qquad [\because j^2 = -1]$$

$$= \pm \omega_1 j$$

$$P = -\frac{\gamma}{2} \pm \omega_1 j$$

from this,

$$P_1 = -\frac{\gamma}{2} + j\omega_1$$

$$P_2 = -\frac{\gamma}{2} - j\omega_1$$

now,

$$x = Be^{pt}$$

$$x = B_1 e^{P_1 t} + B_2 e^{P_2 t}$$

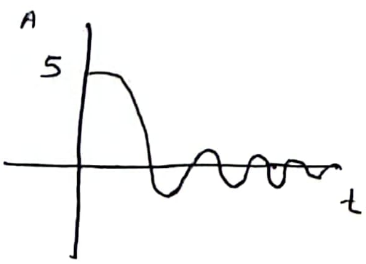
$$= B_1 e^{(-\frac{\gamma}{2} + J\omega_1)t} + B_2 e^{(-\frac{\gamma}{2} - J\omega_1)t}$$

$$= B_1 e^{-\frac{\gamma}{2}t} \cdot e^{J\omega_1 t} + B_2 e^{-\gamma/2 t} \cdot e^{-J\omega_1 t}$$

$$= e^{-\frac{\gamma t}{2}} (B_1 e^{J\omega_1 t} + B_2 e^{-J\omega_1 t})$$

$$= \boxed{A e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \theta)}$$

$y = 5e^{-3x} \cos(3x)$



$A = 5$

$-\frac{\gamma t}{2} = -3x$

$\omega t = 3x$

$t = x$

case-2

$$\omega_0^2 = \frac{\gamma^2}{4}$$

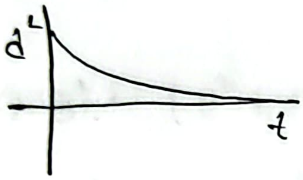
$$P = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2} \qquad \left[\omega_0^2 = \frac{\gamma^2}{4} \right]$$

$$P = -\frac{\gamma}{2}$$

$$x = (A + Bt)\, e^{-\frac{\gamma t}{2}} \qquad \boxed{B e^{-\frac{\gamma t}{2}}} \text{ — ধীরে হবে}$$

→ more appropriate

$$y = 2e^{-5}$$



L-10, 21-10-23

case-3

$$\omega_0^2 < \frac{\gamma^2}{4}$$

$$x = Be^{pt}$$

$$P = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \omega_0^2}$$

$$P = -\frac{\gamma}{2} \pm \lambda$$

$\gamma = \frac{b}{m}$

$\omega_0^2 = \frac{k}{m}$

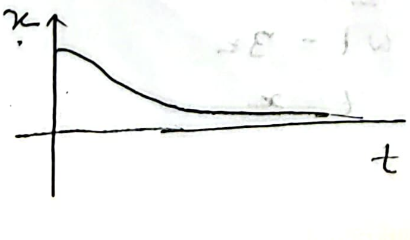
b, k, m all are constant

so, calcolation is constent (λ)

$$P_1 = -\frac{\gamma}{2} + \lambda$$

$$P_2 = -\frac{\gamma}{2} - \lambda$$

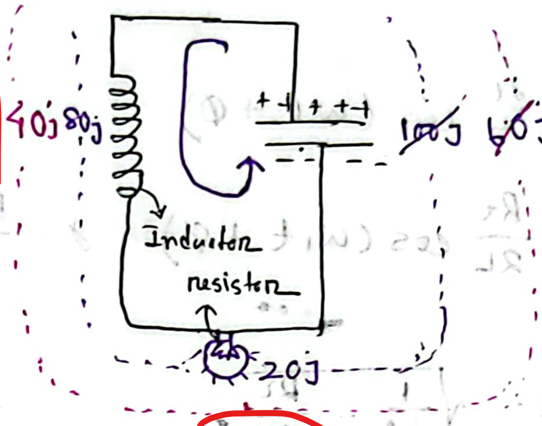
$$\therefore x = B_1 e^{\left(-\frac{\gamma}{2} + \lambda\right)t} + B_2 e^{\left(-\frac{\gamma}{2} - \lambda\right)t}$$



RLC

RLC circuit

$$V_L = L\frac{di}{dt}$$



~~100J~~ ~~60J~~ 20J

এক cycle এ Energy Loss হবে

$$V_c = \frac{Q}{C}$$

$$V = IR$$

$$IR \pm L\frac{di}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{1}{LC}\cdot Q = 0$$

$$\Rightarrow \frac{d^2Q}{dt^2} + \gamma\frac{dQ}{dt} + \omega_0^2 Q = 0$$

ms: $\gamma = \frac{b}{m}$, $\omega_0^2 = \frac{k}{m}$

$\gamma = \frac{R}{L}$, $\omega_0^2 = \frac{1}{LC}$

RLC

$\frac{1}{LC} > \frac{R^2}{4L^2}$

$\frac{1}{LC} = \frac{R^2}{4L^2}$

$\frac{1}{LC} < \frac{R^2}{4L^2}$

mass spring

$\frac{k}{m} > \frac{b^2}{m^2}$

$\frac{k}{m} = \frac{b^2}{m^2}$

$\frac{k}{m} < \frac{b^2}{m^2}$

Mass spring	LC circuit	RC Circuit
$\frac{k}{m} > \frac{l^2}{4m^2}$	$\omega_0^2 > \frac{R^2}{4}$	$\frac{1}{LC} > \frac{R^2}{4L^2}$
$\frac{k}{m} = \frac{l^2}{4m^2}$	$\omega_0^2 = \frac{R^2}{4}$	$\frac{1}{LC} = \frac{R^2}{4L^2}$
$\frac{k}{m} < \frac{l^2}{4m^2}$	$\omega_0^2 < \frac{R^2}{4}$	$\frac{1}{LC} < \frac{R^2}{4L^2}$

Parash

Relational Box

case 1 2 ব্যবহৃত

ω_0 = Resonence Frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\gamma = \frac{R}{L}$$

$$\omega_1^2 = \omega_0^2 - \frac{\gamma}{4}$$

case 1

$$\frac{1}{LC} > \frac{R^2}{4L^2}$$

$$Q(t) = Ae^{-\frac{\gamma t}{2}} \cos(\omega t + \phi)$$

$$Q(t) = Ae^{-\frac{Rt}{2L}} \cos(\omega_1 t + \phi) \; ; \; \gamma = \frac{R}{L}$$

where,

$$\omega_1 = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

if Angular frequency in damping

Linear Resonence Frequency

$$\boxed{f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}}$$

Linear frequency of oscillation

$$\boxed{f_1 = \frac{1}{2\pi}\left(\sqrt{\frac{1}{\sqrt{LC}} - \frac{R^2}{4L^2}}\right)}$$

* $\boxed{\text{Oscillatory}}$ না হলে frequency বের করা যায় না

Example-1

$C = 1\mu F = 10^{-6} F$

$L = 0.2 H$

$R = 800 \Omega \quad f_1 = ?$

$$\frac{1}{LC} = \frac{1}{1\times10^{-6}\times0.2} = 5\times10^{-6}$$

$$\frac{R^2}{4L^2} = \frac{(800)^2}{4\times(0.2)^2} = 4\times10^{6}$$

so, $\frac{1}{LC} > \frac{R^2}{4L^2} \mid \omega_0^2 > \frac{\gamma}{4}^2$

so, it is $\boxed{\text{Oscillatory}}$

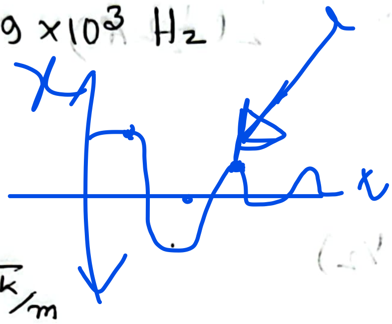
$$\therefore f_1 = \boxed{\frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}}$$

$$= \frac{1}{2\pi}\sqrt{5\times10^6 - 4\times10^6}$$

$$= 0.159\times10^3 \ Hz$$

$$\omega_1 = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

$$2\pi f_1 = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$



book - 15.6

(a) if $b << \sqrt{k/m}$

$$SHM = \omega_0^2 = \frac{k}{m}$$

$$DHM = \omega_0 = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

undamped $\Rightarrow \omega = \frac{2\pi}{T}$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

NO-4

(b) DHM,

$$x(t) = Ae^{-\frac{\gamma t}{2}}$$

$$\frac{A}{2} = Ae^{-\frac{bt}{2m}}$$

$$\Rightarrow \frac{1}{2} = e^{-\frac{bt}{2m}}$$

$$\Rightarrow \ln\left(\frac{1}{2}\right) = -\frac{bt}{2m}$$

$$\Rightarrow t = -\frac{2m}{b}\ln\left(\frac{1}{2}\right)$$

(c) Energy,

$$E_p = \frac{1}{2}kx^2$$

$$= \frac{1}{2}k\left(Ae^{-\frac{bt}{2m}}\right)^2$$

$$= \frac{1}{2}kA^2e^{-\frac{bt}{2m}}$$

initial,

$$E = \frac{1}{2}kA^2$$

$$\frac{1}{2}k\left(Ae^{-\frac{bt}{2m}}\right)^2 = \frac{1}{2}\left(\frac{1}{2}kA^2\right)$$

$$\Rightarrow \left(\frac{1}{2}kA^2\right)e^{-\frac{bt}{2m}} = \frac{1}{2}\left(\frac{1}{2}kA^2\right)$$

module

$$\Rightarrow e^{-\frac{bt}{2m}} = \frac{1}{2}$$

15.5

$$\Rightarrow t = -\frac{b}{m}\ln(1/2)$$

30-10-23

C-11

SHM -

$\omega_0^2 = \frac{k}{m}$

$\gamma = \frac{b}{m}$

$\frac{\omega_0}{\gamma} = 10$

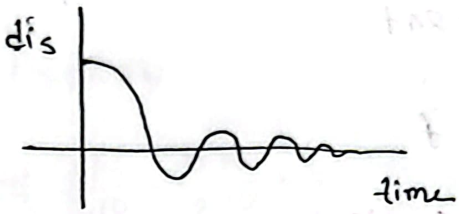
$\omega_0 = 10\gamma$

$\omega_0^2 = 100\gamma^2$

$\omega_0^2 = 400 \cdot \frac{\gamma^2}{4}$

satisfy the condition of Oscillatoriy

so



(4) chapter -15 example, Exercise (1-4)

Waves :

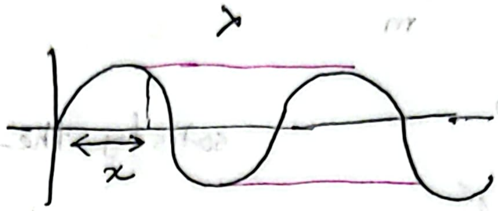
what is waves ?

- longitudinal waves (sound)
- Travelling wave

Weve : transfer energy from one to ~~~~ another wave

$v \rightarrow$ wave velocity

$\lambda \rightarrow$ wave length



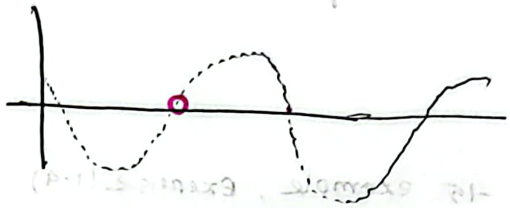
weves speed,

$\boxed{v = f\lambda}$

$\omega = 2\pi f$

$T = \frac{1}{f}$

Progressive wave



$\boxed{y_0 = A \sin \omega t}$

$y_P = A \sin(\omega t - \phi)$

$y = A \sin(\omega t - \frac{2\pi}{\lambda} x)$

$= A \sin(\frac{2\pi}{T} t - \frac{2\pi}{\lambda} x)$

$= A \sin 2\pi (\frac{t}{T} - \frac{x}{\lambda})$

$= A \sin 2\pi (\frac{vt}{\lambda} - \frac{x}{\lambda})$

$\boxed{y = A \sin \frac{2\pi}{\lambda} (vt - x)}$

$$\boxed{\begin{aligned} \text{Phase diff} &= \frac{2\pi}{\lambda} \times \text{path diff} \\ \phi &= \frac{2\pi}{\lambda} \times x \end{aligned}}$$

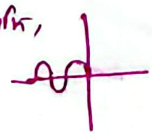
$v = f\lambda$ $\qquad \omega = 2\pi / T$

$\Rightarrow v = \frac{1}{T} \lambda$

$\Rightarrow T = \frac{\lambda}{v}$

show the the equation of the displacement

ধরি,



$y = A \sin(\omega t + \phi)$

$y = A \sin \frac{2\pi}{\lambda} (vt + x)$

Exercise 1

6. $y = 4\sin\pi(0.10x - 2t)$

we know,

$$y = A\sin\frac{2\pi}{\lambda}(vt - x)$$

$$\therefore y = -4\sin\pi(2t - 0.10x)$$

$$= -4\sin\frac{2\pi}{2}(2t - 0.10x)$$

$$= -4\sin\frac{2\pi \times 1}{2}\left(\frac{2t}{0.10} - \frac{0.10}{0.10}x\right)$$

$$= -4\sin\frac{2\pi}{\frac{2}{0.10}}(20t - x)$$

$\therefore$ (i) wave length $= 20\,m$

(ii) speed $v = 20\,ms^{-1}$

(iii) frequency $f = \frac{v}{\lambda} = \frac{20}{20} = 1\,Hz.$

" sin এর ভিতরের কিছু একসাথে ভাগ করা যায় না "

Example 2

$$y = 10\sin\frac{2\pi}{100}(3600t - 20)$$

(i) $A = 10\,m$

(ii)

Progressive wave এর

Differential equation for wave motion

/ Question: show that particle velocity = wave velocity × strain

$$y = A\sin\frac{2\pi}{\lambda}(vt - x)$$

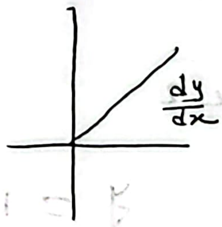
$$\frac{dy}{dx} = A\cos\frac{2\pi}{\lambda}(vt-x)\,\frac{d}{dx}\left(\frac{2\pi}{\lambda}(vt-x)\right)$$

$$\boxed{\frac{dy}{dx} = -\frac{2\pi}{\lambda}A\cos\frac{2\pi}{\lambda}(vt - x)}$$

$\frac{dy}{dx}$ = negative হলে, compression

$\frac{dy}{dx}$ = positive হলে, streater

$$\frac{dy}{dt} = +\frac{2\pi}{\lambda}vA\cos\frac{2\pi}{\lambda}(vt - x)$$



$$\boxed{\therefore \frac{dy}{dt} = -v\frac{dy}{dx}}$$

particle velocity = −wave velocity × strain

V